

WALMART SALES ANALYTICS

End-to-End Data Analysis Report

Revenue Performance | Customer Behavior | Branch Intelligence

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EXECUTIVE SUMMARY

This report presents a comprehensive end-to-end analysis of 9,969 Walmart sales transactions spanning 100 branches across 98 U.S. cities from 2019 to 2023. Using Python for data cleaning, exploratory data analysis (EDA), and feature engineering, and MySQL for business-critical SQL queries, this analysis surfaces the key revenue drivers, operational inefficiencies, and strategic opportunities within Walmart's retail network.

The primary business problem investigated was:

"Which Walmart branches are experiencing the steepest year-over-year revenue decline from 2022 to 2023, by how much, and what operational patterns drive this decline?"

KEY PERFORMANCE INDICATORS

\$1.21M Total Revenue Across 9,969 txns	\$476K Total Profit 39.4% avg margin	\$121.35 Avg Txn Value Median: \$88	5.83 / 10 Avg Rating 35.4% rate ≥7	100 Branches Across 98 cities
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1. DATASET OVERVIEW & DATA QUALITY

1.1 Dataset Profile

Attribute	Detail
Source	Walmart Transactional Sales Data
Raw Records	10,051 rows
After Cleaning	9,969 rows
Duplicates Removed	82 records (0.82%)
Date Range	January 2019 – December 2023
Branches Covered	100 unique branches

Attribute	Detail
Cities Covered	98 unique U.S. cities
Features	11 columns (invoice_id, branch, city, category, unit_price, quantity, date, time, payment_method, rating, profit_margin)

1.2 Data Cleaning Pipeline

The following systematic pipeline was applied to transform raw data into analysis-ready format:

Step	Action Taken	Records Affected	Outcome
1	Removed duplicates using <code>df.drop_duplicates()</code>	82 rows removed	Clean unique transactions
2	Handled missing values via <code>df.dropna()</code>	Null rows dropped	No missing data
3	Stripped '\$' from <code>unit_price</code> ; cast to float	All 9,969 rows	Numeric calculations enabled
4	Parsed 'date' column with <code>pd.to_datetime()</code>	All 9,969 rows	Datetime operations unlocked
5	Loaded cleaned data to MySQL via SQLAlchemy	Full dataset	SQL queries enabled

2. FEATURE ENGINEERING

Four new business-critical features were engineered from existing columns, dramatically expanding analytical capability:

Feature Created	Formula / Method	Business Purpose
total	<code>unit_price × quantity</code>	Calculate actual transaction revenue (missing in raw data)
month	<code>df['date'].dt.month</code>	Enable monthly sales trend and seasonality analysis
hour	<code>pd.to_datetime(df['time']).dt.hour</code>	Identify peak shopping hours for staffing optimization
shift (SQL)	<code>CASE WHEN hour<12 THEN 'morning'...</code>	Classify transactions by time-of-day for operational analysis
year	<code>df['date'].dt.year</code>	Enable year-over-year revenue comparison (primary query)
profit	<code>total × profit_margin</code>	Quantify actual profit contribution per transaction

 **Business Impact:** Without engineering the 'total' field, no revenue analysis was possible. Without 'hour' and 'shift', time-based patterns were invisible. These 6 features directly enabled all 8 SQL business queries.

3. EXPLORATORY DATA ANALYSIS — KEY INSIGHTS

3.1 Revenue & Profit by Category

Fashion Accessories and Home & Lifestyle are the two revenue powerhouses, collectively generating 80.9% of total company revenue — yet they rank last in customer satisfaction.

Category	Revenue	% of Total	Profit	Avg Rating	Rating Rank
Fashion Accessories	\$489,481	40.5%	\$192,315	5.78	#5 of 6
Home and Lifestyle	\$489,250	40.4%	\$192,214	5.74	#6 of 6
Electronic Accessories	\$78,175	6.5%	\$30,772	5.91	#4 of 6
Food & Beverages	\$53,471	4.4%	\$21,553	7.11	#1 of 6
Sports & Travel	\$52,498	4.3%	\$20,614	6.92	#2 of 6
Health & Beauty	\$46,851	3.9%	\$18,672	7.00	#3 of 6

 **Business Impact:** The top 2 revenue categories (81% of revenue) have the LOWEST customer satisfaction scores — 5.74 and 5.78 out of 10. This is a critical vulnerability: low satisfaction in your highest-revenue lines risks long-term customer churn.

 **Recommendation:** Immediately audit product quality, store display, and post-purchase experience for Fashion Accessories and Home & Lifestyle. A 0.5-point improvement in ratings for these two categories could protect and grow the \$978K revenue base they represent. Consider launching a category-specific NPS survey within 30 days.

3.2 Transaction Distribution & High-Value Customer Segment

Metric	Value	Implication
Average Transaction Value	\$121.35	Healthy basket size overall
Median Transaction Value	\$88.00	Median is 27% below mean — right skew confirmed
Top 10% Transaction Threshold	\$222.00	Customer spending \$222+ is a high-value segment
Revenue from Top 10% Customers	\$372,638	30.8% of all revenue from just 10% of transactions
Outlier Transactions (>\$309)	401 transactions	4% of txns but 18.2% of total revenue (\$219,569)
Avg Quantity per Transaction	2.36 items	Low basket size — upselling opportunity

 **Business Impact:** Just 401 outlier transactions (4% of all txns) generate \$219,569 — 18.2% of total revenue. If these customers are lost, it creates an immediate \$220K hole in revenue.

✔ **Recommendation:** Create a VIP/loyalty program targeting the top 10% spenders (transactions ≥\$222). Identify these customers, offer personalized promotions, and focus retention on this segment. Even retaining 50% more of these customers could add \$37K-\$55K annually. Also focus on increasing average basket size — current avg of 2.36 items leaves significant upsell opportunity.

3.3 Payment Method Analysis

Payment Method	Transactions	Revenue	Revenue Share	Avg Txn Value
Credit Card	4,256 (42.7%)	\$488,821	40.4%	\$114.86
Ewallet	3,881 (38.9%)	\$457,316	37.8%	\$117.84
Cash	1,832 (18.4%)	\$263,589	21.8%	\$143.88

💡 **Business Impact:** Cash customers have the HIGHEST average transaction value (\$143.88 — 25% higher than credit card users), yet represent only 18.4% of transactions. This is a missed revenue segment. Additionally, 78.2% of revenue now flows through digital payment channels — a significant operational dependency.

✔ **Recommendation:** Investigate why cash customers spend more per visit. They may be less price-sensitive or making planned bulk purchases. Design targeted outreach or in-store promotions for cash-paying segments. For digital payment infrastructure, ensure Ewallet and Credit Card systems have 99.9% uptime — a system outage now risks 78% of daily revenue.

3.4 Shift & Hour Analysis — Operational Patterns

Shift	Transactions	Revenue	Revenue Share	Avg Txn Value
Afternoon (12pm–5pm)	4,636 (46.5%)	\$578,818	47.8%	\$124.85
Evening (6pm–11pm)	3,246 (32.6%)	\$379,402	31.4%	\$116.88
Morning (Before 12pm)	2,087 (20.9%)	\$251,507	20.8%	\$120.51

Top 5 revenue-generating hours: 3PM (\$142,017) | 4PM (\$134,918) | 7PM (\$128,581) | 5PM (\$116,301) | 6PM (\$113,073)

💡 **Business Impact:** 3PM–5PM is the single most critical sales window, generating \$393,236 (32.5% of all revenue) in just 2 hours. The morning shift generates 20.8% of revenue with 20.9% of traffic — operating at near-full efficiency. Evening is under-monetized relative to traffic volume.

✔ **Recommendation:** Peak-time strategy: Deploy maximum staff and ensure full inventory availability between 3PM–5PM daily. Run 'Happy Hour'-style promotions from 6PM–8PM to boost the underperforming evening segment. Consider reduced hours on low-traffic morning slots at bottom-performing branches to cut operating costs.

3.5 Day-of-Week Analysis

Day	Transactions	vs. Monday (Baseline)	Strategic Note
Tuesday	1,468	+146 (+11.0%)	Consistently highest traffic day
Thursday	1,463	+141 (+10.7%)	Second busiest — steady mid-week peak
Wednesday	1,459	+137 (+10.4%)	Strong mid-week pattern
Saturday	1,441	+119 (+9.0%)	Weekend peak — expected
Sunday	1,411	+89 (+6.7%)	Strong weekend
Friday	1,405	+83 (+6.3%)	Pre-weekend pickup
Monday	1,322	Baseline	Consistently slowest day

 **Business Impact:** Monday consistently has 11% fewer transactions than Tuesday — the weakest day of the week. This represents ~146 missed transactions every Monday, roughly \$17,700 in potential weekly revenue gap.

 **Recommendation:** Launch a 'Monday Boost' promotional campaign — branch-specific discounts or loyalty points multiplier on Mondays. If Monday traffic can be raised to Tuesday-level, the potential lift is approximately \$920,000 annualized across all 100 branches. Even a 5% lift = ~\$46,000 in incremental annual revenue.

3.6 Correlation & Relationship Analysis

Key findings from the correlation heatmap and scatter analysis:

- unit_price vs. total: Strong positive correlation (as expected). Price is the primary lever for transaction value.
- quantity vs. total: Moderate positive correlation. Increasing items per basket is the secondary revenue lever.
- rating vs. total: Near-zero correlation ($r \approx 0.01$). Customer satisfaction has NO meaningful relationship with transaction spend.
- profit_margin varies significantly by category — Fashion and Home have high margins but lower ratings.

 **Business Impact:** The near-zero correlation between rating and spending means Walmart cannot rely on customer satisfaction alone to drive revenue. Happy customers spend the same as neutral ones. Revenue growth must come from price strategy, basket size, and traffic volume — not just satisfaction improvements.

4. SQL BUSINESS QUERIES — INSIGHTS & IMPACT

4.1 PRIMARY FINDING: Branch Revenue Decline (2022 vs. 2023)

This query — the core business question — identified branches with the highest year-over-year revenue deterioration using CTEs and DENSE_RANK window functions:

Branch	2022 Revenue	2023 Revenue	Revenue Lost	Decline %	Priority
WALM045	\$1,731	\$647	-\$1,084	-62.6%	CRITICAL
WALM047	\$2,581	\$1,069	-\$1,512	-58.6%	CRITICAL
WALM098	\$2,446	\$1,030	-\$1,416	-57.9%	CRITICAL
WALM033	\$2,099	\$931	-\$1,168	-55.7%	HIGH
WALM081	\$1,723	\$850	-\$873	-50.7%	HIGH

CRITICAL FINDING: 5 branches lost an average of 57.1% of their revenue in just one year. WALM045 lost 62.6% — nearly two-thirds of its business. Total revenue lost across these 5 branches: \$6,053. If left unaddressed and extended to more branches, this pattern could represent \$60,000+ in systemic risk across the network.

Contrast with the top 5 growing branches — showing the performance gap is widening:

Branch	2022 Revenue	2023 Revenue	Revenue Gained	Growth %
WALM006	\$765	\$2,088	+\$1,323	+172.9%
WALM010	\$808	\$2,114	+\$1,306	+161.6%
WALM091	\$962	\$2,394	+\$1,432	+148.9%
WALM072	\$811	\$1,785	+\$974	+120.1%
WALM077	\$862	\$1,884	+\$1,022	+118.6%

 **Recommendation:** Deploy an immediate branch rescue task force for WALM045, WALM047, WALM098, WALM033, and WALM081. Study the playbooks of top-growing branches (WALM006, WALM010, WALM091) and replicate their operational strategies — staffing models, promotional approaches, category mix — at declining branches. Potential revenue recovery: \$3,000–\$5,000 per branch if decline is halted.

4.2 Revenue Concentration — The 80/20 Problem

Branch Segment	Branch Count	Revenue Generated	% of Total Revenue
Top 20 branches	20 (20%)	\$437,828	36.2%
Middle 60 branches	60 (60%)	\$626,055	51.8%
Bottom 20 branches	20 (20%)	\$145,843	12.0%

💡 Business Impact: The bottom 20 branches (WALM092 at \$5,039 to WALM097 at \$7,377) collectively generate only 12% of revenue — less than 1/3 of the top 20's contribution. The worst branch (WALM092: \$5,039) generates just 19.6% of the best branch's revenue (WALM009: \$25,688).

✅ Recommendation: Conduct a financial viability audit on the bottom 10 branches. For branches with consistent multi-year decline and below-\$6,000 annual revenue, evaluate restructuring, merging service areas, or converting to smaller-format stores. Reallocating resources from bottom 5 branches to top-performing ones could yield a 15–20% portfolio efficiency gain.

4.3 Payment Method by Branch — Digital Readiness Gap

Analysis of preferred payment method by branch revealed a critical operational insight: 82 of 100 branches show Ewallet as the preferred method, while 18 branches prefer Credit Card. No branch listed Cash as the preferred method — yet cash drives 21.8% of overall revenue.

💡 Business Impact: Cash is generating 21.8% of revenue but is not the preferred method at any branch — suggesting cash customers are not being actively served or incentivized. The branches with Credit Card preference (like WALM003, WALM009) tend to also be the highest revenue branches.

✅ Recommendation: Investigate whether the 18 credit-card-preferred branches have better POS infrastructure, more affluent catchment areas, or different customer demographics. Standardize Ewallet acceptance quality across all 100 branches — inconsistent digital payment experience can suppress revenue at branches where digital is dominant.

4.4 Highest Rated Category by Branch — Personalization Opportunity

The SQL analysis revealed that top-rated categories vary significantly by branch and city — Health & Beauty leads in cities like Abilene (avg 9.7), Bedford (8.15), and multiple others, while Food & Beverages leads in Austin (9.3), Alice (7.68), and Angleton (7.7).

💡 Business Impact: There is no single 'nationally best' category. Customer satisfaction is hyperlocal — what works in Austin does not work in Amarillo. One-size-fits-all inventory and promotional strategy will systematically underperform.

✅ Recommendation: Implement branch-level category personalization. Allocate 15–20% more shelf space and promotional budget to the locally top-rated category per branch. For Health & Beauty dominant markets (Abilene, Bedford), run targeted wellness promotions. For Food & Beverage dominant markets (Austin, Alice), emphasize freshness and variety.

4.5 Profit by Category — True Business Value

Category	Total Revenue	Total Profit	Profit Margin %	Profit Rank
Fashion Accessories	\$489,481	\$192,315	39.3%	#1
Home and Lifestyle	\$489,250	\$192,214	39.3%	#2
Electronic Accessories	\$78,175	\$30,772	39.4%	#3
Food & Beverages	\$53,471	\$21,553	40.3%	#4
Sports & Travel	\$52,498	\$20,614	39.3%	#5
Health & Beauty	\$46,851	\$18,672	39.9%	#6

 **Business Impact:** Profit margins are nearly identical across all categories (39.3%–40.3%). This means revenue growth = profit growth almost proportionally. There is no 'hidden high-margin' category to pivot toward — the path to profit is purely revenue growth and cost management, not category switching.

 **Recommendation:** Since margins are uniform, focus revenue efforts where volume is highest (Fashion and Home). However, Food & Beverages has the highest margin (40.3%) AND the highest customer satisfaction (7.11) — making it the ideal category for aggressive growth investment. Doubling Food & Beverages revenue from \$53K to \$106K would add \$21.5K in profit at minimal incremental cost.

5. CONSOLIDATED BUSINESS RECOMMENDATIONS

The following 6 recommendations are prioritized by estimated revenue impact and implementation feasibility:

Priority	Recommendation	Target	Est. Revenue Impact	Timeline
#1 — CRITICAL	Branch Rescue Program for top 5 declining branches (WALM045, 047, 098, 033, 081)	Stop -57% avg decline	\$5,000–\$6,000 recovery/yr	Immediate (30 days)
#2 — HIGH	VIP Loyalty Program for top 10% spenders (txn ≥\$222)	Retain high-value segment	+\$37K–\$55K annually	60–90 days
#3 — HIGH	Category quality audit for Fashion Accessories and Home & Lifestyle (rating 5.74–5.78)	Raise ratings to 6.5+	Protect \$978K revenue base	60 days
#4 — MEDIUM	Monday Traffic Campaign — promotions & loyalty multipliers	Close 11% Monday gap	+\$46K annually (5% lift)	30 days
#5 — MEDIUM	Peak-hour staffing optimization (3PM–5PM) and Evening promotions (6PM–8PM)	Maximize 47.8% afternoon share	+\$25K–\$40K annually	45 days
#6 — LONG-TERM	Branch-level category personalization — allocate inventory by locally top-rated category	Improve rating + revenue	+5–10% branch revenue	90–180 days

6. ANALYTICAL CONCEPTS & TECHNIQUES USED

Category	Concept / Tool	Where Applied
Data Pipeline	End-to-end ETL (Extract, Transform, Load)	CSV → Python cleaning → MySQL database
Data Cleaning	Duplicate detection & removal, null handling, type casting	82 duplicates removed; \$-string fixed; datetime parsed
Feature Engineering	Derived column creation from raw fields	total, month, hour, shift, year, profit
Statistical Analysis	Descriptive stats, skewness, IQR, outlier detection	Transaction value distribution; boxplot analysis
Univariate Analysis	Distribution analysis of single variables	Countplot, histplot, boxplot per column
Bivariate Analysis	Relationship between two variables	Category vs. revenue; rating vs. total; payment vs. revenue
Correlation Analysis	Pearson correlation matrix + heatmap	Identified near-zero rating-spend correlation
Outlier Detection	IQR method + boxplots	Found 401 high-value outlier transactions
SQL CTEs	WITH clause for modular multi-step queries	All 8 SQL business queries
SQL Window Functions	RANK(), DENSE_RANK() with PARTITION BY	Busiest day per branch; YoY revenue ranking
SQL Self-Join	Joining same table on different year values	YoY revenue comparison for 2022 vs. 2023
SQL CASE WHEN	Conditional logic for categorical derivation	Shift classification (morning/afternoon/evening)
Python-to-SQL Integration	SQLAlchemy ORM engine for database loading	Pushed cleaned DataFrame into MySQL
Data Visualization	Seaborn + Matplotlib for EDA charts	8+ chart types including heatmap, pairplot, scatter

7. CONCLUSION

This analysis successfully identified and quantified the primary business problem: five Walmart branches are experiencing 50–62% year-over-year revenue collapse, while five other branches are growing at 120–173%. The performance gap is widening — and without intervention, the declining branches risk becoming financially unviable.

Beyond the primary finding, the analysis surfaced six actionable opportunities — from protecting the \$978K revenue base concentrated in two underperforming-in-satisfaction categories, to capitalizing on the high-value customer segment that drives 30.8% of revenue from just 10% of transactions.

The end-to-end approach — from raw CSV through Python EDA to MySQL analytics — demonstrates a complete data analyst workflow: data ingestion, quality validation, feature engineering, exploratory analysis, statistical testing, and SQL-driven business intelligence.

If all 6 business recommendations are implemented,
estimated annual revenue uplift potential:

\$150,000 – \$220,000+

on a \$1.21M revenue base — a 12–18% portfolio performance improvement